Scratch: Move and Control Sprites

Arrow keys, mouse control, bouncing and gliding: the first things every game needs.

Scratch | Beginner | Ages 6–12

The Big Idea

- A sprite is any character or object in your project. Motion blocks (blue) change where it is and which way it faces.
- The stage is 480 wide and 360 tall. The middle is x: 0, y: 0.
- x goes from -240 (left) to 240 (right). y goes from -180 (bottom) to 180 (top).
- Direction is an angle: 0 is up, 90 is right, 180 is down and -90 is left.
- Combine motion blocks with events and loops and you can control anything.

Motion Blocks You Will Use Most

- `move (10) steps` goes forward in the direction the sprite faces.
- `turn right (15) degrees` or `turn left (15) degrees` rotates the sprite.
- `go to x: (0) y: (0)` jumps straight to a spot. `glide (1) secs to x: y:` slides there.
- `change x by (10)` moves sideways and `change y by (10)` moves up or down.
- `set x to` and `set y to` put the sprite at an exact column or row.
- `point in direction (90)` and `point towards [mouse-pointer]` choose where to face.
- `if on edge, bounce` stops a sprite leaving the stage.

Ways to Control a Sprite

- Keyboard: `when [arrow] key pressed` plus `change x by` or `change y by`.
- Mouse: `go to [mouse-pointer]` or `set x to (mouse x)` inside a `forever` loop.
- Automatic: `forever` with `move (3) steps` and `if on edge, bounce`.
- Smooth keys: inside `forever`, use `if <key [right arrow] pressed?> then` so holding a key keeps moving.

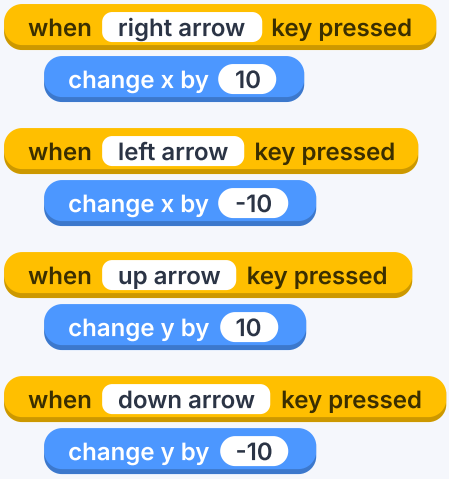
Rotation Style

- Click the sprite's info panel to find Direction and Rotation style.
- All around lets the sprite spin fully. Left-right only flips it so it does not go upside down.
- Don't rotate means it always faces the same way, which is best for platformer characters.

💻 Script Examples

Arrow key controls

Scratch blocks

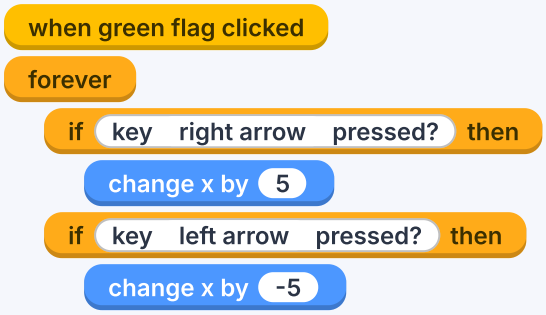


▶ What happens:

Each key press nudges the sprite 10 steps in that direction.

Smooth movement (hold the key)

Scratch blocks

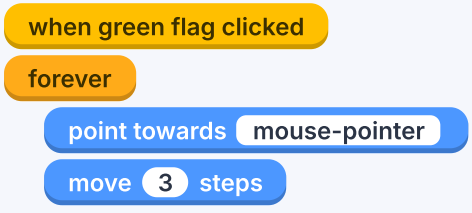


▶ What happens:

Holding a key keeps the sprite moving smoothly, with no pause between steps.

Follow the mouse

Scratch blocks

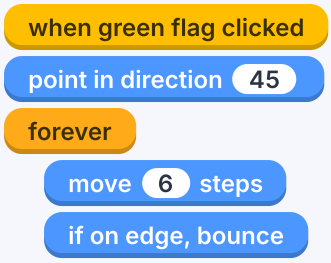


▶ What happens:

The sprite turns toward the pointer and chases it across the stage.

Bouncing ball

Scratch blocks

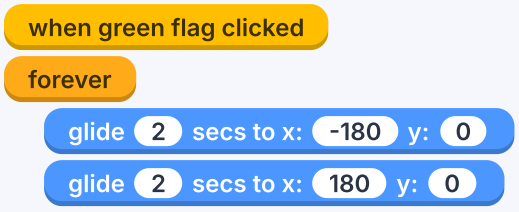


▶ What happens:

The ball travels at an angle and bounces off every edge.

Patrol with glide

Scratch blocks

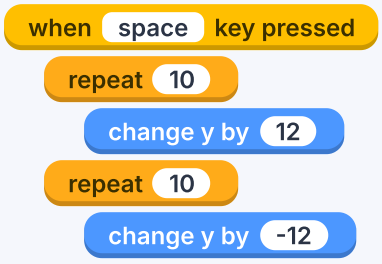


▶ What happens:

A guard slides from the left side to the right side and back again, forever.

A simple jump

Scratch blocks



▶ What happens:

The sprite rises 120 steps and falls back to where it started.

📚 Key Terms

Sprite
A character or object in your project.

Stage
The area where your project runs: 480 by 360.

x position
How far left or right. 0 is the middle.

y position
How far up or down. 0 is the middle.

Direction
Which way the sprite faces, in degrees.

Glide
Move smoothly to a spot over some seconds.

Bounce
Turn around when touching the edge.

Coordinates
The pair (x, y) that names a spot on the stage.

Rotation style
How a sprite is allowed to turn.

Event
Something like a key press that starts a script.

🧮 Quick Reference

```
Stage: x -240..240, y -180..180
```

```
Directions: 0 up, 90 right, 180 down,
-90 left
```

```
move (10) steps
```

```
turn right (15) degrees
```

```
go to x: (0) y: (0)
```

```
glide (1) secs to x: (100) y: (50)
```

```
change x by (10) / change y by (10)
```

```
if on edge, bounce
```

✏️ Worked Examples

Q: Make a sprite walk right when you press the right arrow.

A: Step 1: Add when [right arrow] key pressed. Step 2: Snap change x by (10) under it. Step 3: Click the green flag and press the key.

Q: Make a ball bounce around forever.

A: Step 1: when green flag clicked. Step 2: point in direction (45). Step 3: forever containing move (6) steps and if on edge, bounce.

Q: Put a sprite exactly in the top left corner.

A: Use go to x: (-240) y: (180). The top left of the stage is the smallest x and the largest y.

🚫 Common Mistakes

- Mixing up x and y. x is sideways and y is up and down.
- Using `move` with a negative number and wondering why it goes backwards.
- Forgetting `forever` or the key check, so the sprite only moves once.
- Letting the sprite rotate upside down. Set Rotation style to Left-right.
- Starting without `go to`, so the sprite begins in a different place each time.

👨‍👩‍👧 Parent Corner

- Play Treasure Hunt: hide an object in the room and give your child x and y style directions like 3 steps forward, turn right.
- Ask them to explain what each arrow key block does in their own words.
- Challenge them: Can you make the sprite stay inside a box?
- This is the maths of coordinates and angles. Ask which direction is 90.
- Ask them to change one number and predict the result before testing.

🎯 Quick Check

1. What is the x and y of the middle of the stage?

✅ x: 0 and y: 0.

2. Which direction is 90 degrees?

✅ Right.

3. Which block makes a sprite turn around at the edge?

✅ if on edge, bounce.

4. How do you move a sprite up 10 steps?

✅ change y by (10).

5. Which blocks make smooth movement while a key is held?

✅ A forever loop with if <key pressed?> then and change x by.